Data dictionary

Supplied fields

Field	Use	Meaning / caveat
`code`	Audit key	Barcode/code supplied as text; historical labels may have lost leading zeroes.
`product_name`	Descriptive	May be missing or noisy; not a predictive input.
`brands`	Descriptive	Brand string; not a predictive input.
`categories` , `categories_fr`	Descriptive	Source category labels; no guaranteed legal scope.
`categories_tags`	Scope audit	Exact tags used for the taxonomy-recognized sensitivity subset.
`main_category_fr`	Descriptive	Main category shown in EDA; inherited metadata can be inaccurate.
`countries_tags`	Scope audit	Multi-valued market/country tags; not exclusive shares.
`ingredients_text`	Quality check	Raw INCI/transcription text, sometimes with OCR or label prose.
`ingredients_list`	Supplied feature source	Python-literal ingredient list; parsed with `ast.literal_eval` .
`n_ingredients`	Audit / feature source	Supplied count, checked against the re-parsed count.
`has_risk`	Supervised target	Supplied binary proxy; exact 16-substance labeling registry is unavailable.

Engineered audit and split fields

Field	Type / role	Construction
`parsed_ingredients`	List / CSV JSON string	Normalized ingredient entries; no invented synonym mapping.
`ingredient_count`	Integer / EDA and model input source	Number of non-empty parsed entries.
`formula_group`	Split key	SHA-256 fingerprint of the sorted unique normalized ingredient set.
`scope_status`	Categorical / sensitivity	Recognized, unclassified, or explicitly excluded category.
`category_label`	Descriptive	Main category or `Unknown`.
`raw_list_overlap`	Float / audit	Fraction of list tokens found in raw source text.
`low_alignment_flag`	Boolean / quarantine	Raw/list overlap below 0.2; a quality heuristic, not clinical validation.
`long_list_flag`	Boolean / audit	Ingredient count above Q3 + 1.5×IQR; retained.
`long_prose_flag`	Boolean / audit	Any parsed entry above 200 characters; retained.

Random Forest model matrix

The final model learns its columns within each training fold: the top **1,500 normalized ingredients** become binary presence features, and `log1p(ingredient_count)` is added as a numeric feature. Categories, brands, product names, country tags, and external deterministic registry matches are not inputs. The model does not use TF-IDF features. A positive prediction is not a safety determination.